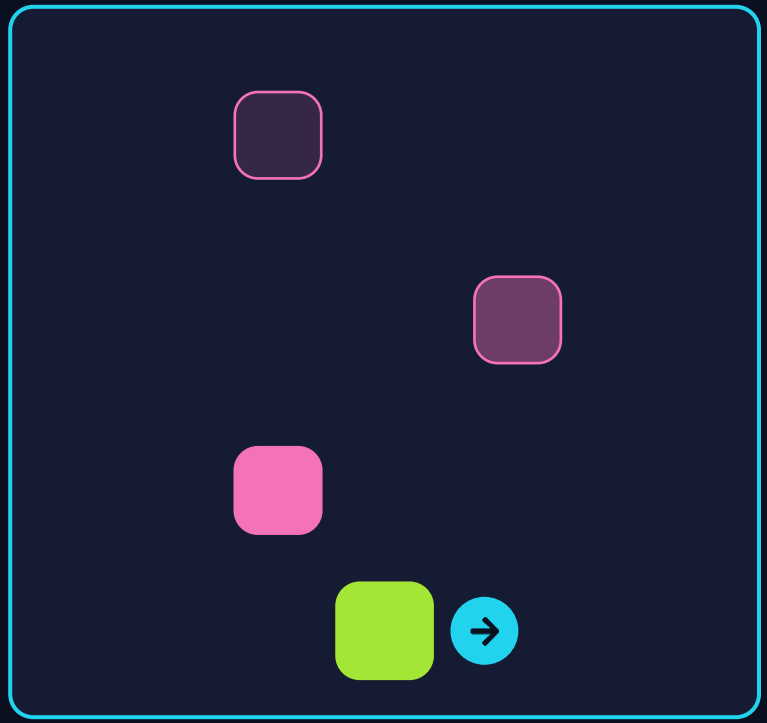
CCA GAME DEVELOPMENT

Dodge the Block

Module 2 — Session 8

Teacher: Annisa Hendrawan

Junior High School — Global Prestasi School



```
block_y += 5
```

Today's Goal

By the end of this session:

01



Combine movement and keyboard controls

02



Make blocks fall using X and Y coordinates

03



Keep the player inside the screen

04



Build a simple survival game

Can You Dodge the Falling Blocks?

Remember Session 5, 6, and 7? Now, let's combine everything!



SESSION 5 Move objects using X and Y.



SESSION 6 Control the player using the keyboard.



SESSION 7 Keep the player inside the screen.

OUR MISSION

Move the player left and right.

Avoid the falling blocks.

Survive for as long as possible!

MOVE + KEYBOARD + BOUNDARIES = SURVIVE

How Can We Move an Object Down?

Remember the coordinate system?

X → horizontal

Y → vertical

To make a block fall, we increase its Y position.

```
block_y += 5
```

Every time this code runs, the block moves down by 5 pixels.



Make the Block Move Automatically

The game loop repeats while the game is running.

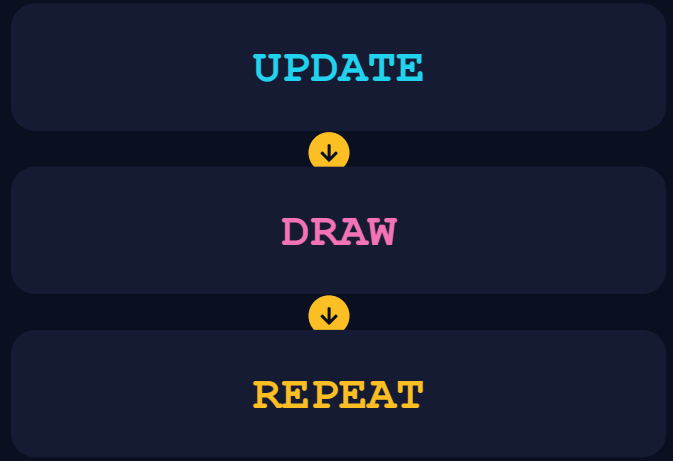
```
while running:

    # Handle events

    block_y += block_speed

    # Draw objects

    pygame.display.flip()
    clock.tick(60)
```



Remember: Movement code must be inside the game loop to create continuous movement.

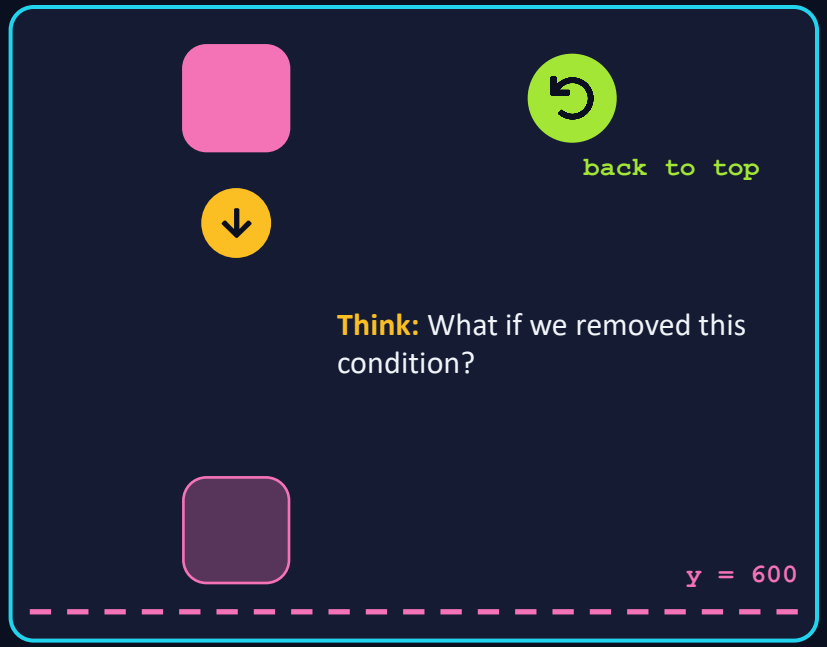
What Happens at the Bottom?

The block keeps falling until it leaves the screen. How can we make it return to the top?

```
if block_y > 600:
    block_y = 0
```

CHECK → **RESET**

If Y becomes greater than 600, the block returns to the top. Then it starts falling again!



Mission 1: Build Dodge the Block

BUILD STEP BY STEP

1. Create the player near the bottom of the screen.
2. Add LEFT and RIGHT keyboard controls.
3. Add boundaries to keep the player inside the screen.
4. Create a falling block.
5. Make the block move down automatically.
6. Make the block return to the top after falling past the bottom.



Session 8 starter code
`github.com/annisafzhn/CCA.git`

TEST YOUR GAME

- Can you move the player?
- Does the block fall?
- Does the player stay inside the screen?
- Does the block return to the top?

Success condition: The player can move while the block falls automatically.

Mission 2: Test Your Game Logic

Complete these challenges in order.

1 Change the Speed

```
block_speed = 10
```

What happens to the falling block?

A. It moves more slowly.

B. It moves faster.

C. It stops moving.

Run the game and check your answer.

2 Change the Starting Position

Change block_x from 300 to 600:

```
300 → 600
```

Where will the block appear?

3 Debug the Code

Find the problem:

```
while running:
  # Other game code
  if block_y > 600:
      block_y = 0
  # Other game code
```

The block falls, but it never seems to move.

Hint: Is the code that increases block_y inside the game loop?

Fix the code and test your game!

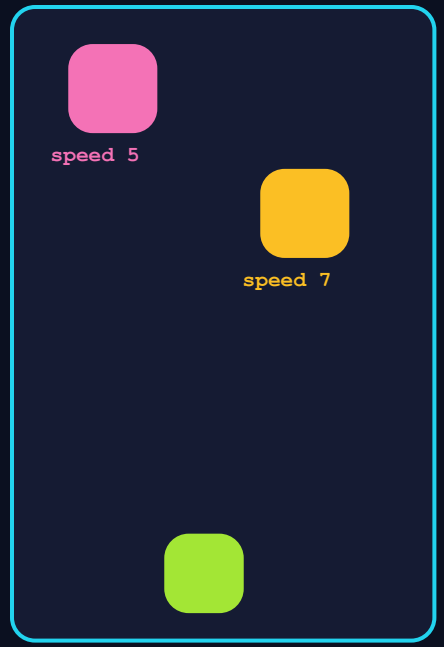
Bonus: Add Another Falling Block

Your game currently has one falling block. Can you make it more challenging?

```
block2_x = 500
block2_y = 100
block2_speed = 7
```

Your mission: Create a second block with its own position and speed. Make it fall independently.

Bonus goal: Make the game harder without breaking the player movement!



THINK

Can both blocks fall at different speeds?

Can the player dodge both blocks?

How does adding another block change the difficulty?

Today's Big Ideas

X AND Y COORDINATES
Control an object's position on the screen.

KEYBOARD INPUT
Lets the player move left and right.

BOUNDARIES
Keep the player inside the screen.

GAME LOOP
Updates and draws objects repeatedly.

FALLING MOVEMENT
Increasing Y makes a block move down.

IF STATEMENT
Checks whether a condition is true.

NEXT SESSION: LOADING ASSETS (IMAGES)

Module 3 — Session 9

Today we used rectangles. Next, we replace basic shapes with real images and backgrounds!

Can you turn your simple game into something that looks like a real game? See you in Session 9! 👋